AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

# TM105
# Introduction to Computer Programming

## Midterm Examination (MTA)
## Fall Semester 2025/2026

## Date: 15/Nov./2025

| | |
|---|---|
| **Number of Exam Pages:** 4 (including this cover sheet) | **Time Allowed:** 1.5 hours |

**Instructions:**

- Total Marks: **60**
- This exam consists of 5 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.



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## Question 1 [6 Marks]

For each question, choose the most appropriate answer. Copy the answer to the External Answer booklet. Each question carries 2 marks.

1. Which of the following is a valid class name in Java?
   a. short
   b. Short
   c. CLASS!
   d. Both a and b are valid

2. Which of the following is not a literal?
   a. "33"
   b. 1+3
   c. 2
   d. 33.2

3. What is the value of the Boolean expression: !(10 != 9)
   a. true
   b. 9
   c. false
   d. 10

## Question 2 [12 marks]

The following code includes some errors. Find the errors and rewrite the code after correction.

```
public class MTA {
    public static void main(String args) {
        int x = "10";
        string text = "Positive";

        // Check if x is positive and print the result
        if x > 0
            System.out.println("text");
        Else
            System.out.println("Negative");
    }
}
```

## Question 3 [15 Marks]

Trace the following pieces of code and write their exact output:

a.

```
int x= 2;
int y = 11;
double z = 2;
System.out.println(y/x);
System.out.println(y/z);
System.out.println(y+x*z);
System.out.println((y+x)*z);
```



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b.

```
for (int a=1;a<=10;a+=3){
  System.out.println(a);
  a++;
}
```

c.

```
int x = 40;
int y = 40;
if (x>y && x !=10)
{
    System.out.println(x);
    System.out.println(y);
}
else
{
   x+=y;
   y-=2;
   System.out.println(x);
   System.out.println(y);
}
```

d.

```
for (int i = 9; i > 5; i--)
  if (i%2==0)
      System.out.println(i*2);
  else
      System.out.println(i);
```

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For questions 4 and 5, copy the below code into your answer sheet and continue the rest of the code according to the given specifications of each question.

```
import java.util.Scanner;
public class MTA {
     public static void main(String[] args) {

    }
}
```

## Question 4 [20 Marks]:

The monthly household water usage cannot exceed 500 liters. The usage category is determined as follows:

| Usage category | Water Usage |
|---|---|
| Low | number of liters <= 100 |
| Medium | 100 < number of liters ≤ 250 |
| High | 250 < number of liters ≤ 400 |
| Very High | 400 < number of liters ≤ 500 |

You need to write a Java program that works as follows:



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- Read from the user an integer that represents the number of liters of water used.
- Find and print the water usage category based on the above table.
- If the user entered a value of number of liters that is negative or more the 500, the program should print an error message.

Here are different samples of program run to help you understand how the program works:

| | |
|---|---|
| 1 | Enter the number of liters of water used: 80<br>Low |
| 2 | Enter the number of liters of water used: 200<br>Medium |
| 3 | Enter the number of liters of water used: 400<br>High |
| 4 | Enter the number of liters of water used: 480<br>Very High |
| 5 | Enter the number of liters of water used: 520<br>Invalid number of liters! |

## Question 5 [7 Marks]:

Write a for loop to print the multiples of 5 from 50 down to 5 in one line.

The output should be exactly as follows:

50 45 40 35 30 25 20 15 10 5

**End of questions**



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